Unit III: Communication Networking Techniques

❖ Circuit Switching

In circuit switching network resources (bandwidth) are divided into pieces and bit delay is constant during a connection. The dedicated path/circuit established between sender and receiver provides a guaranteed data rate. Data can be transmitted without any delays once the circuit is established.

Drawbacks:

Inefficient use of resources: Circuit switching requires the establishment of a dedicated communication path between two nodes, which means that the resources along that path, such as bandwidth and switch ports, are reserved for the duration of the communication. This can result in inefficient use of resources, as the resources may remain unused during periods of low or no communication.
Limited scalability: Circuit switching is not well-suited for large-scale networks with many nodes, as it requires a dedicated communication path between each pair of nodes. This can result in a high degree of complexity and difficulty in managing the network.
Vulnerability to failures: Circuit switching relies on a dedicated communication path, which can make the network vulnerable to failures, such as cable cuts or switch failures. In the event of a failure, the communication path must be re-established, which can result in delays or loss of data.
Delay and latency: Circuit switching requires the establishment of a dedicated communication path, which can result in delay and latency in establishing the path and transmitting data. This can impact the real-time performance of applications, such as voice and video.
High cost: Circuit switching requires the reservation of resources, which can result in a high cost, particularly in large-scale networks. This can make circuit switching less practical for some applications.
Lack of flexibility: Circuit switching is not flexible as it only allows one type of communication at a time, such as voice or data. This can limit the ability of users to perform multiple tasks simultaneously.
Limited mobility: Circuit switching is not well-suited for mobile devices or nodes that move frequently, as it requires the establishment of a dedicated communication path. This can result in communication disruptions or dropped calls.
Limited capacity: Circuit switching can have limited capacity as it requires the establishment of a dedicated communication path between two nodes. This can limit the number of simultaneous communications that can occur.
High setup time: Circuit switching requires a significant setup time to establish the dedicated communication path between two nodes. This can result in delays in initiating communication.
No prioritization: Circuit switching does not provide any mechanism for prioritizing certain types of traffic over others. This can result in delays or poor performance for time-critical applications, such as voice or video.

Advantages of Circuit Switching:

1. The main advantage of circuit switching is that a committed transmission channel is established between the computers which give a guaranteed data rate.
2. In-circuit switching, there is no delay in data flow because of the dedicated transmission path.
3. **Reliability:** Circuit switching provides a high level of reliability since the dedicated communication path is reserved for the entire duration of the communication. This ensures that the data will be transmitted without any loss or corruption.
4. **Quality of service:** Circuit switching provides a guaranteed quality of service, which means that the network can prioritize certain types of traffic, such as voice and video, over other types of traffic, such as email and web browsing.
5. **Security:** Circuit switching provides a higher level of security compared to packet switching since the dedicated communication path is only accessible to the two communicating parties. This can help prevent unauthorized access and data breaches.
6. **Ease of management:** Circuit switching is relatively easy to manage since the communication path is pre-established and dedicated to a specific communication. This can help simplify network management and reduce the risk of errors.
7. **Compatibility:** Circuit switching is compatible with a wide range of devices and protocols, which means that it can be used with different types of networks and applications. This makes it a versatile technology for various industries and use cases.

Disadvantages of Circuit Switching:

It has the following disadvantages :

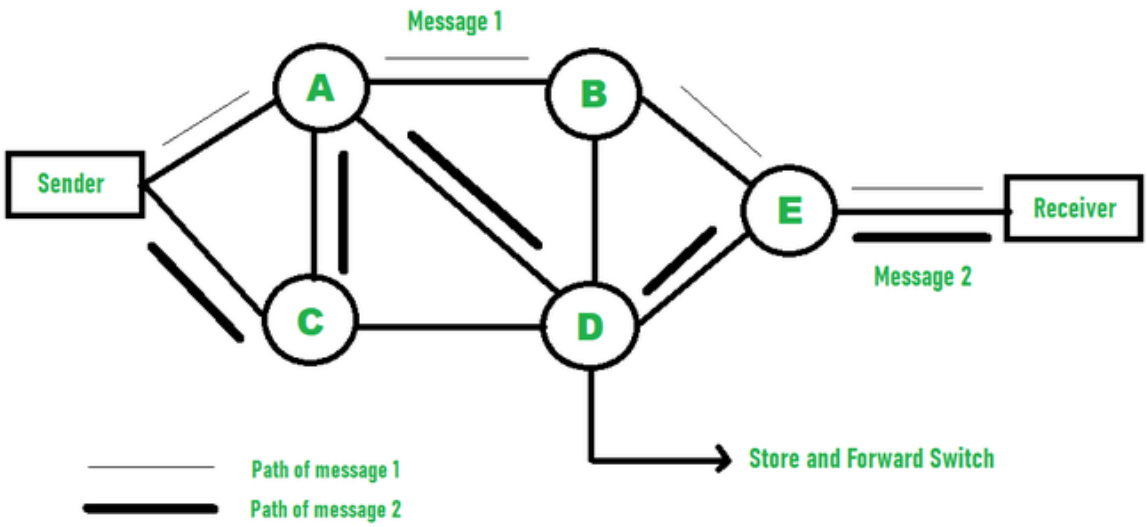
1. It takes a long time to establish a connection.
2. More bandwidth is required in setting up dedicated channels.
3. It cannot be used to transmit any other data even if the channel is free as the connection is dedicated to circuit switching.
4. **Limited Flexibility:** Circuit switching is not flexible as it requires a dedicated circuit between the communicating devices. The circuit cannot be used Waste of Resources for any other purpose until the communication is complete, which limits the flexibility of the network.
5. **Waste of Resources:** Circuit switching reserves the bandwidth and network resources for the duration of the communication, even if there is no data being transmitted. This results in the wastage of resources and inefficient use of the network.
6. **Expensive:** Circuit switching is an expensive technology as it requires dedicated communication paths, which can be costly to set up and maintain. This makes it less feasible for small-scale networks and applications.
7. **Susceptible to Failure:** Circuit switching is susceptible to failure as it relies on a dedicated communication path. If the path fails, the entire communication is disrupted. This makes it less reliable than other networking technologies, such as packet switching.
8. **Not suitable for bursty traffic:** Circuit switching is not suitable for bursty traffic, where data is transmitted intermittently at irregular intervals. This is because a dedicated circuit needs to be established for each communication, which can result in delays and inefficient use of resources.

❖ **Packet Switching**

Packet Switching in computer networks is a method of transferring data to a network in the form of packets. In order to transfer the file fast and efficiently manner over the network and minimize the transmission latency, the data is broken into small pieces of variable length, called **Packet**. At the destination, all these small parts (packets) have to be reassembled, belonging to the same file. A packet is composed of a payload and various control information. No pre-setup or reservation of resources is needed.
Packet Switching uses the **Store and Forward** technique while switching the packets; while forwarding the packet each hop first stores that packet then forwards. This technique is very beneficial because packets may get discarded at any hop for some reason. More than one path is possible between a pair of sources and destinations. Each packet contains the Source and destination address using which they independently travel through the network. In other words, packets belonging to the same file may or may not travel through the same path. If there is congestion at some path, packets are allowed to choose different paths possible over an existing network.

Diagram of Packet Switching

In packet switching the data is divided into small packets which allow faster movement of data. Each packet contains two parts that is Header and Payload, the header on each packet conation information. Below is the diagram of how packet switching works.



Packet Switching

Types of Delays in Packet Switching

- **Transmission Delay:** Time required by the **spent** station to transmit data to the link.
- **Propagation Delay:** Time of data propagation through the link.
- **Queueing Delay:** Time spent by the packet at the destination's queue.
- **Processing Delay:** Processing time for data at the destination.

Advantages of Packet Switching over Circuit Switching

- More efficient in terms of bandwidth, since the concept of reserving a circuit is not there.
- Minimal transmission latency.
- More reliable as a destination can detect the missing packet.

- More fault tolerant because packets may follow a different path in case any link is down, Unlike Circuit Switching.
- Cost-effective and comparatively cheaper to implement.

Disadvantage of Packet Switching over Circuit Switching

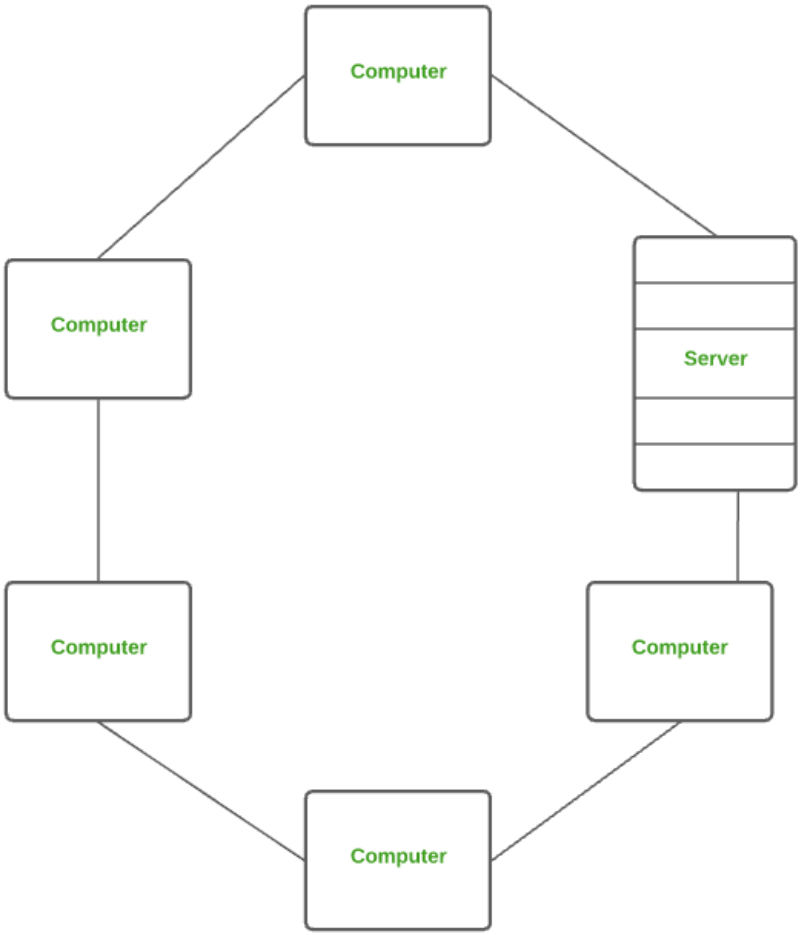
- Packet Switching doesn't give packets in order, whereas Circuit Switching provides ordered delivery of packets because all the packets follow the same path.
- Since the packets are unordered, we need to provide sequence numbers for each packet.
- Complexity is more at each node because of the facility to follow multiple paths.
- Transmission delay is more because of rerouting.
- Packet Switching is beneficial only for small messages, but for bursty data (large messages) Circuit Switching is better.

Difference Between Packet Switching and Circuit Switching

Packet Switching	Circuit Switching
In packet switching data is divided into packets, and packets is sent independently.	There is a dedicated path for each packet in circuit switching.
In Packet switching, data is processed at all intermediate nodes including the source system.	In-Circuit switching, data is processed at the source system only.
The delay between data units in packet switching is not uniform.	The delay between data units in circuit switching is uniform.
Packet switching is less reliable.	Circuit switching is more reliable.
Transmission of the data is done not only by the source but also by the intermediate routers.	Transmission of the data is done by the source.
Less wastage of resources.	Wastage of resources is more in Circuit Switching.
In Packet Switching there is no physical path between the source and the destination.	In-Circuit Switching there is a physical path between the source and the destination.
Call setup is not required in packet switching.	Call setup is required in circuit switching.
Packet switching requires complex protocols for delivery.	Circuit switching requires simple protocols for delivery.
Latency is high in Packet switching.	Latency is low in circuit switching.
Littel bit more overheating in packet switching.	Overheading is low in circuit switching.

LAN (Local Area Network)

LAN stands for **Local-area Network**. It is a computer network that covers a relatively small area such as within a building or campus of up to a few kilometers in size. LANs are generally used to connect personal computers and workstations in company offices to share common resources, like printers, and exchange information. A number of experimental and early commercial LAN technologies were developed in the 1970s. Cambridge Ring was developed at Cambridge University in 1974. Ethernet was developed at Xerox PARC between 1973 and 1974. ARCNET was developed by Data Point Corporation in 1976 and announced in 1977.



How Do LANs Work?

A router serves as the hub where the majority of LANs connect to the Internet. Home LANs often utilise a single router, but bigger LANs may also use network switches to transmit packets more effectively.

LANs nearly always connect devices to the network via Ethernet, WiFi, or both of these technologies. Ethernet is a physical network connection protocol that calls for Ethernet cables. WiFi is a protocol for using radio waves to connect to a network.

Servers, desktop computers, laptops, printers, Internet of Things (IoT) devices, and even game consoles can connect to LANs. LANs are frequently used in offices to give internal staff members shared access to servers or printers that are linked to the network.

Types of LAN

There are mainly two types of LANs exist.

- **Client/Server LANs:** Multiple devices (the clients) are connected to a main server in a client/server LAN. The server controls network traffic, device access, application access, and file storage. Any connected device that runs apps or accesses the Internet qualifies as a client. Clients can use wired or wireless connections to connect to the server.
- **Peer-to-Peer LANs:** Peer-to-peer LANs are often smaller because they lack a central server and can't support huge workloads like client/server LANs can. Every device on a peer-to-peer LAN contributes equally to the network's operation. Through wired or wireless connections to a switch or router, the devices share data and resources. Peer-to-peer networks are the norm in homes.

What is a Virtual LAN?

The same physical network can have its traffic divided into two networks using virtual LANs, or VLANs. Imagine establishing two independent LANs in the same room, each with its own router and Internet connection. Similar to that, but with only one router and one Internet connection required, VLANs divide networks virtually rather than physically.

VLANs are beneficial for network management, particularly in very large LANs. Administrators may much more simply control the network by segmenting it. (Subnets, another method of segmenting networks for increased efficiency, differ greatly from VLANs.)

What are the Benefits of a LAN?

- **Privacy:** LAN is a private network, thus no outside regulatory body controls it, giving it a privacy.
- **High Speed:** LAN offers a much higher speed(around 100 mbps) and data transfer rate comparatively to WAN.
- **Supports different transmission mediums:** LAN support a variety of communications transmission medium such as an Ethernet cable (thin cable, thick cable, and twisted pair), fiber and wireless transmission.
- **Inexpensive and Simple:** A LAN usually has low cost, installation, expansion and maintenance and LAN installation is relatively easy to use, good scalability.

What are the Drawback of LAN?

- The initial setup costs of installing Local Area Networks is high because there is special software required to make a server.
- Communication devices like an ethernet cable, switches, hubs, routers, cables are costly.
- LAN administrator can see and check personal data files as well as Internet history of each and every LAN user. Hence, the privacy of the users are violated
- LANs are restricted in size and cover only a limited area
- Since all the data is stored in a single server computer, if it can be accessed by an unauthorized user, can cause a serious data security threat.

Network Protocols

What is Network Protocol?

A network protocol is a set of rules that govern data communication between different devices in the network. It determines what is being communicated, how it is being communicated, and when it is being communicated. It permits connected devices to communicate with each other, irrespective of internal and structural differences.

How do Network Protocols Work?

It is essential to understand how devices communicate over a network by recognizing network protocols. The Open Systems Interconnection (OSI), the most widely used model, illustrates how computer systems interact with one another over a network. The communication mechanism between two network devices is shown by seven different layers in the OSI model. Every layer in the OSI model works based on different network protocols. At every layer, one or more protocols are there for network communication. To enable network-to-network connections, the Internet Protocol (IP), for instance, routes data by controlling information like the source and destination addresses of data packets. It is known as a network layer protocol.

Types of Network Protocols

In most cases, communication across a network like the Internet uses the OSI model. The OSI model has a total of seven layers. Secured connections, network management, and network communication are the three main tasks that the network protocol performs. The purpose of protocols is to link different devices.

The protocols can be broadly classified into three major categories:

1. Network Communication
2. Network Management
3. Network Security

1. Network Communication

Communication protocols are really important for the functioning of a network. They are so crucial that it is not possible to have computer networks without them. These protocols formally set out the rules and formats through which data is transferred. These protocols handle syntax, semantics, error detection, synchronization, and authentication. Below mentioned are some network communication protocol:

- **Hypertext Transfer Protocol(HTTP)**

It is a layer 7 protocol that is designed for transferring a hypertext between two or more systems. HTTP works on a client-server model, most of the data sharing over the web is done through using HTTP.

- **Transmission Control Protocol(TCP)**

TCP layouts a reliable stream delivery by using sequenced acknowledgment. It is a connection-oriented protocol i.e., it establishes a connection between applications before sending any data. It is used for communicating over a network. It has many applications such as emails, FTP, streaming media, etc.

- **User Datagram Protocol(UDP)**

It is a connectionless protocol that lay-out a basic but unreliable message service. It adds no flow control, reliability, or error-recovery functions. UPD is functional in cases where reliability is not required. It is used when we want faster transmission, for multicasting and broadcasting connections, etc.

- **Border Gateway Protocol(BGP)**

BGP is a routing protocol that controls how packets pass through the router in an independent system one or more networks run by a single organization and connect to different networks. It connects the endpoints of a LAN with other LANs and it also connects endpoints in different LANs to one another.

- **Address Resolution Protocol(ARP)**

ARP is a protocol that helps in mapping logical addresses to the physical addresses acknowledged in a local network. For mapping and maintaining a correlation between these logical and physical addresses a table known as ARP cache is used.

- **Internet Protocol(IP)**

It is a protocol through which data is sent from one host to another over the internet. It is used for addressing and routing data packets so that they can reach their destination.

- **Dynamic Host Configration Protocol(DHCP)**

it's a protocol for network management and it's used for the method of automating the process of configuring devices on IP networks. A DHCP server automatically assigns an IP address and various other configurational changes to devices on a network so they can communicate with other IP networks. it also allows devices to use various services such as NTP, DNS, or any other protocol based on TCP or UDP.

2. Network Management

These protocols assist in describing the procedures and policies that are used in monitoring, maintaining, and managing the computer network. These protocols also help in communicating these requirements across the network to ensure stable communication. Network management protocols can also be used for troubleshooting connections between a host and a client.

- **Internet Control Message Protocol(ICMP)**

It is a layer 3 protocol that is used by network devices to forward operational information and error messages. ICMP is used for reporting congestions, network errors, diagnostic purposes, and timeouts.

- **Simple Network Management Protocol(SNMP)**

It is a layer 7 protocol that is used for managing nodes on an IP network. There are three main components in the SNMP protocol i.e., SNMP agent, SNMP manager, and managed device. SNMP agent has the local knowledge of management details, it translates those details into a form that is compatible with the SNMP manager. The manager presents data acquired from SNMP agents, thus helping in monitoring network glitches, and network performance, and troubleshooting them.

- **Gopher**

It is a type of file retrieval protocol that provides downloadable files with some description for easy management, retrieving, and searching of files. All the files are arranged on a remote computer in a stratified manner. Gopher is an old protocol and it is not much used nowadays.

- **File Transfer Protocol(FTP)**

FTP is a Client/server protocol that is used for moving files to or from a host computer, it allows users to download files, programs, web pages, and other things that are available on other services.

- **Post Office Protocol(POP3)**

It is a protocol that a local mail client uses to get email messages from a remote email server over a TCP/IP connection. Email servers hosted by ISPs also use the POP3 protocol to hold and receive emails intended for their users. Eventually, these

users will use email client software to look at their mailbox on the remote server and to download their emails. After the email client downloads the emails, they are generally deleted from the servers.

- **Telnet**

It is a protocol that allows the user to connect to a remote computer program and to use it i.e., it is designed for remote connectivity. Telnet creates a connection between a host machine and a remote endpoint to enable a remote session.

3. Network Security

These protocols secure the data in passage over a network. These protocols also determine how the network secures data from any unauthorized attempts to extract or review data. These protocols make sure that no unauthorized devices, users, or services can access the network data. Primarily, these protocols depend on encryption to secure data.

- **Secure Soket Layer(SSL)**

It is a network security protocol mainly used for protecting sensitive data and securing internet connections. SSL allows both server-to-server and client-to-server communication. All the data transferred through SSL is encrypted thus stopping any unauthorized person from accessing it.

- **Hypertext Transfer Protocol(HTTPS)**

It is the secured version of HTTP. this protocol ensures secure communication between two computers where one sends the request through the browser and the other fetches the data from the web server.

- **Transport Layer Security(TLS)**

It is a security protocol designed for data security and privacy over the internet, its functionality is encryption, checking the integrity of data i.e., whether it has been tampered with or not, and authentication. It is generally used for encrypted communication between servers and web apps, like a web browser loading a website, it can also be used for encryption of messages, emails, and VoIP.

❖ System Network Architecture (SNA)

Systems Network Architecture (SNA) is a data communication architecture established by IBM to specify common conventions for communication among the wide array of IBM hardware and software data communication products and other platforms. Among the platforms that implement SNA in addition to mainframes are IBM's Communications Server on Windows, AIX, and Linux, Microsoft's Host Integration Server (HIS) for Windows, and many more.

A networking application is dependent on the communication protocol it uses. Every protocol provides an application programming interface (API). TCP/IP's API is called *socket programming* and SNA has its own API. Migrating a networking application from one protocol to another (that is, from SNA to TCP/IP) requires replacing the calls to the API. Business managers are reluctant to invest in protocol conversion only for the sake of changing the underlying protocol without introducing new functions and improvements.